PROFESSIONAL EDUCATION



SUMMER 2023

MACHINE LEARNING FOR BIG DATA AND TEXT PROCESSING: FOUNDATIONS

DATES	DURATION	FORMAT	FEE	CEUS
June 5–6, 2023	2 Days	On Campus	$2,500	1.2

Acquire the expertise you need to immediately implement new strategies for driving value in your organization

Machine learning is a rapidly expanding area with a diverse collection of tools and approaches. Successfully applying such methods to real tasks may seem to require expertise that many do not possess. However, all these methods share the same basic concepts, use the same building blocks.

Understanding these basics, formulations, and when they are appropriate, is key to using machine learning techniques successfully in practice. This foundational course covers the essential concepts and methods in machine learning, providing participants with an entry level expertise they need to get started and quickly move ahead.

EARN A PROFESSIONAL CERTIFICATE

This course may be taken individually or as part of the ***Professional Certificate Program in Machine Learning & Artificial Intelligence***.

WHO SHOULD ATTEND

- **Data and software engineers** who need to build up their foundational machine learning toolkit
- **Computer scientists, programmers, and developers** who are looking to master a diverse set of machine learning tools and approaches for solving practical workplace challenges
- **Data scientists** who want to refresh their knowledge of foundational machine learning concepts, formulations, and algorithms
- **Executives, managers, and other leaders** who need a deeper understanding of how machine learning can improve their organization's operations
- **Project Managers** who want to explore the basics of machine learning and discover how these methods can be used to solve business problems

professional.mit.edu/mlbd

LEARNING OUTCOMES

- Understand the basic machine learning concepts and methods including neural networks
- Learn how to formulate/set up problems as machine learning tasks
- Assess which types of methods are likely to be useful for a given class of problems
- Understand strengths and weakness of learning algorithms

Teaching Methodology

The certificate curriculum is grounded in the spirit of MIT's motto, "Mens et Manus," or "mind and hand," which combines theoretical instruction with hands-on methods of discovery.



INTERACTIVE LECTURES



HANDS-ON EXERCISES



GROUP PROJECTS



SUPPLEMENTAL RESOURCES

LEAD INSTRUCTORS

Regina Barzilay



Regina Barzilay is a Delta Electronics professor in the Department of Electrical Engineering and Computer Science and a member of the Computer Science and Artificial Intelligence Laboratory at the Massachusetts Institute of Technology.

Tommi Jaakkola



Tommi Jaakkola is a professor of Electrical Engineering and Computer Science and also a member of the Computer Science and Artificial Intelligence Laboratory. His work pertains to inferential, algorithmic and estimation questions in machine learning, including large scale probabilistic distributed inference, deep learning, and causal inference.

Stefanie Jegelka



Stefanie Jegelka is an X-Consortium Career Development Assistant Professor in the Department of Electrical Engineering and Computer Science at MIT, where she is a member of CSAIL, and affiliated with IDSS.

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